MATHEMATICS / INDEPENDENT STUDY

MM5020

Mathematics III - Abstract Algebra

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1 Unit 1 · Groups and cyclic structure

1.1 Lecture 1 · From operations to groups

Learning goals and prerequisites
Recognize a group, prove elementary identities, and distinguish a definition from a property. Prerequisites: functions, composition, integers, and proof by contradiction.

Recall and approach. Recall associativity permits reassociation but not reordering.

A symmetry is an action that can be performed and reversed. Rotating a square twice is again a symmetry; doing nothing is a symmetry; reversing a rotation is possible. These three observations suggest associativity, an identity, and inverses. Commutativity is deliberately absent: a reflection followed by a rotation need not equal the reverse order.

Definition 1 · Group
A group is a set G with an operation $G \times G \to G$, written $(a, b) \mapsto ab$, such that $(ab)c = a(bc)$, there is e with $ea = ae = a$, and every a has a^{-1} with $aa^{-1} = a^{-1}a = e$. It is **abelian** if $ab = ba$ for all a, b.

The operation is part of the structure. The integers form a group under addition, but not multiplication: 2 has no integer multiplicative inverse. Nonzero rational numbers form a multiplicative group. In S_X, the bijections of a set X, the operation is composition. Composition is associative because both $(fg)h$ and $f(gh)$ send x to $f(g(h(x)))$. The inverse of a bijection provides the inverse required by the group axioms.

Theorem 1 · Identity, inverses, and cancellation
A group has a unique identity and a unique inverse for each element. If $ab = ac$, then $b = c$. Also $(ab)^{-1} = b^{-1}a^{-1}$.

Proof strategy. Uniqueness proofs compare two candidates. Cancellation uses multiplication by an inverse, with parentheses justified by associativity. **Proof.** If e, e' are identities, then $e = ee' = e'$. If b, c are inverses of a, then $b = b(ac) = (ba)c = c$. From $ab = ac$, multiply on the left by a^{-1} and reassociate to obtain $b = c$. Finally $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = e$, and the reverse product is also e. Uniqueness identifies this as the inverse. □

Example 1 · Solving without commuting
In G, the equation $axb = c$ has the unique solution $x = a^{-1}cb^{-1}$. Multiplying on the correct side matters. The expression $ca^{-1}b^{-1}$ need not work. Substitution verifies existence; cancellation proves uniqueness.

For an integer n, define $a^0 = e$, positive powers by repeated multiplication, and $a^{-n} = (a^{-1})^n$. Induction gives $a^{m+n} = a^m a^n$ for integers m, n. This law concerns powers of one element; it does not justify $(ab)^2 = a^2b^2$ in an arbitrary group.

Watch for this. Closure is a condition on the chosen set. Nonzero matrices are not a group under multiplication, since a nonzero matrix can be singular. The invertible matrices $\mathrm{GL}_{n(K)}$ are a group.

Example 2 · A calculation that detects commutativity
Suppose $(ab)^2 = a^2b^2$ for all a, b in a group. Expanding gives $abab = aabb$. Multiply on the left by a^{-1} and on the right by b^{-1} to get $ba = ab$. Thus this familiar-looking square rule, if universally valid, actually forces the group to be abelian. The converse follows by commuting the middle factors. This is a useful test of an algebraic shortcut: ask what additional property it would imply if it held for every pair.

To present a group example well, name the set and operation, then verify closure before inverses. Writing a symbolic inverse is insufficient if it lies outside the chosen set.

Checkpoint 1.1. Why does $ab = e$ imply $ba = e$ in a group?

Takeaways. Prove structural facts from the axioms before using them as algebraic shortcuts. Study guide U1 revisits cancellation; exercises U1.01–U1.04 test the basic definitions.
Continue with study guide U1 and exercise sheet U1.

1.2 Lecture 2 · Subgroups, generators, and cyclic groups

Learning goals and prerequisites
Use the subgroup test, identify generated subgroups, and classify subgroups of cyclic groups. Recall divisibility and the integer division algorithm.

Recall and approach. Recall the least positive integer principle and Bézout coefficients for a gcd.

Some symmetries preserve additional information. Rotations of a square form a group inside all its symmetries. A subgroup packages this idea without requiring another operation.

Definition 2 · Subgroup and generated subgroup
A subgroup $H \leq G$ is a nonempty subset which is a group under the restricted operation. For $S \subseteq G$, the subgroup $\langle S \rangle$ generated by S is the intersection of all subgroups containing S. Equivalently it consists of finite products of elements of S and their inverses, including the empty product e.

Proof strategy. For a subgroup test, recover identity and inverses first. For cyclic structure, use the least positive exponent and division.

Theorem 2 · One-step subgroup test
A nonempty subset $H \subseteq G$ is a subgroup if and only if $ab^{-1} \in H$ for every $a, b \in H$.

Proof. Necessity follows from closure and inverses. Conversely choose $h \in H$. Then $hh^{-1} = e \in H$; for $a \in H$, $ea^{-1} = a^{-1} \in H$. Applying the condition to a and b^{-1} gives $ab \in H$. Associativity is inherited from G. □

The intersection construction really is a subgroup: the identity belongs to every participating subgroup, and membership of a, b in each gives membership of ab^{-1} in each. The finite-word description follows because those words form a subgroup containing S, and every subgroup containing S contains every such word.

Definition 3 · Order and cyclicity
The order of a, written $\operatorname{ord}(a)$, is the smallest positive n with $a^n = e$, if such n exists; otherwise it is infinite. A group is cyclic if it equals $\langle a \rangle$ for some a.

Theorem 3 · Subgroups of a cyclic group
Every subgroup of $\langle a \rangle$ is cyclic. If a has finite order n, then $\operatorname{ord}\left(a^k\right) = \frac{n}{\gcd(n,k)}$.

Proof strategy. Find the least positive exponent in a subgroup and use division with remainder. For the order formula, translate the problem into divisibility. **Proof.** The trivial subgroup is generated by e. Otherwise choose the least positive d with $a^d \in H$; such a d exists by taking inverses when necessary. For $a^m \in H$, divide $m = qd + r$ with $0 \le r < d$. Then $a^r = a^{m\left(a^d\right)^{-q}} \in H$, so minimality forces $r = 0$. Thus $H = \langle a^d \rangle$. For finite order, $\left(a^k\right)^t = e$ exactly when n divides kt. Writing $n = gn'$ and $k = gk'$ with $g = \gcd(n,k)$ and $\gcd(n',k') = 1$ shows the least positive t is n'. □

Example 3 · Generators modulo twelve
In the additive group $\mathbb{Z}/12\mathbb{Z}$, the order of $[8]$ is $\frac{12}{\gcd(12,8)} = 3$. Its subgroup is $\{[0],[8],[4]\}$. The generators of the whole group are $[1],[5],[7],[11]$, precisely the residues relatively prime to 12.

Watch for this. A subgroup of a cyclic group is cyclic, but a subgroup of a general abelian group need not be generated by one element. In $C_2 \times C_2$, every nonidentity element has order two.

Example 4 · Finding a subgroup from two generators
In $\frac{\mathbb{Z}}{30}\mathbb{Z}$, the subgroup generated by $[12]$ and $[18]$ equals the subgroup generated by $[6]$. Both twelve and eighteen are multiples of six, so every generated element is in the latter subgroup. Conversely $[6] = [18] - [12]$ is generated by the first two, proving equality. Its order is five, and its elements are $[0],[6],[12],[18],[24]$.

The two-inclusion proof generalizes: a subgroup generated by residues a, b is controlled by $\gcd(n,a,b)$. A list of several initial multiples suggests an answer, but the inclusions explain why no later product or inverse produces an overlooked element.

Checkpoint 1.2. Describe every subgroup of the additive group $\mathbb{Z}$.

Takeaways. Generators turn structural questions into arithmetic, but only after the relevant group and operation have been fixed. Complete U1.05–U1.12 before moving to cosets.
Continue with study guide U1 and exercise sheet U1.

2 Unit 2 · Cosets and permutations

2.1 Lecture 3 · Cosets and Lagrange's theorem

Learning goals and prerequisites
Construct cosets, distinguish left from right cosets, and use divisibility of orders correctly. Recall equivalence relations and partitions.

Recall and approach. Recall an equivalence relation partitions a set and a bijection preserves cardinality.

For integers, congruence modulo n groups numbers into translates of $n\mathbb{Z}$. Cosets extend this construction to arbitrary groups. They are sets of elements, not automatically elements of another group.

Definition 4 · Left coset and index
For $H \leq G$, the left coset of H represented by g is $gH = \{gh : h \in H\}$. The index $[G : H]$ is the number of distinct left cosets.

Proof strategy. Prove coset equality by a membership criterion, then count a disjoint partition.

Theorem 4 · Coset partition
Two left cosets are equal or disjoint. Moreover $aH = bH$ if and only if $b^{-1}a \in H$, and each coset has the same cardinality as H.

Proof. If $ah = bk$, then $b^{-1}a = kh^{-1} \in H$. Conversely if $a = bu$ with $u \in H$, then $aH = buH = bH$, because $uH = H$. Every g belongs to gH, so the cosets cover G, and the preceding argument proves disjointness. Left multiplication $h \mapsto gh$ is a bijection $H \to gH$, with inverse $x \mapsto g^{-1}x$. □

Theorem 5 · Lagrange
If G is finite and $H \leq G$, then $|G| = [G : H]|H|$. In particular the order of an element divides $|G|$.

Proof. Count the elements in the disjoint coset partition. Apply the result to $\langle a \rangle$, which has $\operatorname{ord}(a)$ elements. □

Consequently $a^{|G|} = e$ for finite G. A group of prime order p is cyclic: any $a \neq e$ has order dividing p and greater than one, hence order p. These are consequences of a theorem, not extra assumptions about the operation.

Example 5 · A nonnormal subgroup already has cosets
In S_3, let $H = \{e, (12)\}$. There are three left cosets, each containing two permutations. The left coset $(13)H$ need not equal the right coset $H(13)$. Lagrange's theorem uses neither commutativity nor normality.

The converse to Lagrange is false in general: a divisor of a group order need not occur as a subgroup order. We will give a concrete example using A_4 after discussing normality. When using Lagrange, record the direction: an existing subgroup imposes divisibility; divisibility alone does not construct a subgroup.

Watch for this. The statement $a^{|G|} = e$ does not mean that a has order $|G|$. In $C_2 \times C_2$, all elements satisfy $a^4 = e$, although no element has order four.

Example 6 · Left and right cosets visibly differ
In S_3 with $H = \{e, (12)\}$, compute $(13)H = \{(13), (123)\}$ using right-to-left composition. On the other side, $H(13) = \{(13), (132)\}$. Both sets contain two elements and both contain (13), but they are not equal. This does not violate the left-coset partition theorem, which compares two left cosets of the same subgroup, not a left coset with a right coset.

When solving an order problem, write "divides" before choosing among possible values. The theorem restricts candidates; a construction or another theorem must establish that a candidate actually occurs.

Checkpoint 2.1. Can a group of order 35 have an element of order 6?

Takeaways. Cosets are a counting tool before they become an algebraic construction. U2.01–U2.06 practice translating between membership, equality, and counting.
Continue with study guide U2 and exercise sheet U2.

2.2 Lecture 4 · Permutations, cycle structure, and parity

Learning goals and prerequisites
Compute compositions, determine orders from disjoint cycles, and prove that parity is well defined. Prerequisites: bijections and determinants.

Recall and approach. Recall composition order and that swapping two columns changes a determinant sign.

Permutations are a concrete laboratory for noncommutative algebra. We compose right to left: in $\sigma\tau$, apply τ first. A cycle $(a_1 a_2 \ldots a_r)$ sends each listed element to the next and the last to the first, fixing all others.

Proof strategy. Follow each orbit to obtain disjoint cycles, then use an invariant determinant to prove parity is independent of factorization.

Theorem 6 · Disjoint-cycle decomposition
Every permutation of a finite set is a product of disjoint cycles, unique up to reordering cycles and rotating each cycle's notation. Its order is the least common multiple of the cycle lengths.

Proof. Start at an element and repeatedly apply the permutation. Finiteness forces a repetition; injectivity ensures the first repeated element is the starting element. This gives one cycle. Repeat on the remaining elements. The resulting orbits are disjoint and determined by the permutation, proving uniqueness in the stated sense. A power fixes every element exactly when its exponent is divisible by every cycle length, proving the order formula. □

Every cycle is a product of transpositions:

$$(a_1 a_2 \ldots a_r) = (a_1 a_r)(a_1 a_{r-1}) \ldots (a_1 a_2). \tag{1}$$

The order of these transpositions can be checked by following a_1, an intermediate a_i, and a_r separately.

Definition 5 · Sign and alternating group
Let P_σ be the matrix with $P_\sigma e_i = e_{\sigma(i)}$. Define $\operatorname{sgn}(\sigma) = \det(P_\sigma)$. The alternating group is $A_n = \{\sigma \in S_n : \operatorname{sgn}(\sigma) = 1\}$.

Theorem 7 · Parity
The sign map is a homomorphism $S_n \to \{1, -1\}$. Every transposition has sign -1, so any two transposition expressions for the same permutation have the same parity. For $n \geq 2$, $|A_n| = \frac{n!}{2}$.

Proof. Matrix multiplication gives $P_{\sigma\tau} = P_\sigma P_\tau$, and determinants multiply. A transposition interchanges two basis columns, so its determinant is -1. An expression with r transpositions therefore has sign $(-1)^r$, independent of the expression because the determinant is fixed. Multiplication by a fixed transposition bijects even and odd permutations. There are $n!$ permutations in total. □

Example 7 · Two independent computations
The permutation $(142)(35)$ has order $\operatorname{lcm}(3, 2) = 6$ and sign $(-1)^2(-1) = -1$. Order and sign answer different questions; an odd permutation can have an order much larger than two.

Watch for this. Disjoint cycles commute. Arbitrary cycles do not. A factorization into transpositions is highly nonunique even though its parity is fixed.

Example 8 · Two descriptions of the same permutation
For $\sigma = (1234)(13)$, tracking one gives $1 \to 4 \to 1$, while tracking two gives $2 \to 3 \to 2$. Thus $\sigma = (14)(23)$, with order two and positive sign. Independently its sign is the product of the negative signs of the four-cycle and transposition, again positive. These two calculations check different aspects of the answer.

An efficient checking procedure is to list the image of each letter once, close its cycle, and remove those letters from consideration. Never infer the order of a product by multiplying the orders of non-disjoint factors; here orders four and two produced order two, not eight.

Checkpoint 2.2. What are the sign and order of a four-cycle?

Takeaways. Cycle decomposition supports both computation and proof. The determinant supplies a conceptual reason that parity does not depend on a chosen factorization.
Continue with study guide U2 and exercise sheet U2.

3 Unit 3 · Homomorphisms and quotients

3.1 Lecture 5 · Kernels, images, and normality

Learning goals and prerequisites
Use a homomorphism to compare groups, identify its kernel and image, and characterize normal subgroups. Recall functions, inverse images, and cosets.

Recall and approach. Recall inverse images and that normality is invariance under every conjugation.

A useful map between groups must respect their operations. Reduction modulo n, the determinant, and the sign of a permutation look different but all translate one group calculation into another.

Definition 6 · Homomorphism
A map $f : G \to H$ is a homomorphism if $f(ab) = f(a)f(b)$. Its kernel is $\ker f = \{g : f(g) = e_H\}$ and its image is $\operatorname{im} f = \{f(g) : g \in G\}$. An isomorphism is a bijective homomorphism.

First prove the automatic identities. The equation $f(e_G) = f(e_G)^2$ and cancellation give $f(e_G) = e_H$. Applying f to $gg^{-1} = e_G$ gives $f(g^{-1}) = f(g)^{-1}$. The inverse function of an isomorphism respects multiplication because applying f verifies the required equation.

Proof strategy. Use the subgroup test for kernel and image, then translate equal outputs into a kernel element.

Theorem 8 · Kernel criterion
The kernel and image of a homomorphism are subgroups. A homomorphism is injective exactly when its kernel is trivial. Its kernel is normal: $g(\ker f)g^{-1} = \ker f$ for every $g \in G$.

Proof. For kernel elements a, b, $f(ab^{-1}) = e$, so the subgroup test applies. For image elements $f(a), f(b)$ their quotient is $f(ab^{-1})$. If $f(a) = f(b)$, then $b^{-1}a \in \ker f$, so a trivial kernel implies $a = b$. Conversely an injective map has only e_G mapping to e_H. Finally $f(gng^{-1}) = f(g)e_H f(g)^{-1} = e_H$. This gives one inclusion for the conjugated kernel; replacing g by g^{-1} gives equality. □

Definition 7 · Normal subgroup
A subgroup $N \leq G$ is normal, written $N \lhd G$, if $gNg^{-1} = N$ for all g. Equivalently, left and right cosets agree: $gN = Ng$.

Example 9 · Three recurring kernels
The kernel of reduction $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is $n\mathbb{Z}$. The kernel of $\det : \mathrm{GL}_{n(K)} \to K^\times$ is $\mathrm{SL}_{n(K)}$. The kernel of sign is A_n. These examples establish normality without enumerating conjugates.

Theorem 9 · Index two
Every subgroup of index two is normal.

Proof. For $g \in H$, both gH and Hg equal H. For $g \notin H$, both are the complement of H: the two left cosets partition G, as do the two right cosets. Thus $gH = Hg$ for all g. □

Watch for this. Being a subgroup is not enough to be a kernel. In S_3, $\{e, (12)\}$ is not normal since $(13)(12)(13)^{-1} = (23)$.

Example 10 · The same image with different kernels
Consider $f : \mathbb{Z} \to \frac{\mathbb{Z}}{6}\mathbb{Z}$ given by $f(n) = [2n]$. Its image is $\{[0], [2], [4]\}$ and its kernel is $3\mathbb{Z}$. It is not surjective onto its stated codomain, but it is surjective onto its image. The quotient predicted by the first isomorphism theorem will therefore have three elements, not six.

For a candidate homomorphism, test the identity and products before calculating a kernel. A subset defined as the inverse image of identity has the useful kernel properties only after multiplication preservation is established. A map with arbitrary formula does not acquire normality just because its zero set was named "kernel."

Checkpoint 3.1. Why can a nontrivial homomorphism $C_p \to H$, for prime p, not identify distinct elements?

Takeaways. The kernel measures loss of information; normality is exactly the closure property needed to turn that loss into a quotient.
Continue with study guide U3 and exercise sheet U3.

3.2 Lecture 6 · Quotients and isomorphism theorems

Learning goals and prerequisites
Construct quotient groups, prove representative independence, and apply the first, second, and third isomorphism theorems.

Recall and approach. Recall normality permits multiplication of cosets and that a kernel detects equality of outputs.

Definition 8 · Quotient group
For $N \lhd G$, the set G/N becomes a group under $(aN)(bN) = abN$. Its identity is N and $(aN)^{-1} = a^{-1}N$.

The key issue is well-definedness. Replacing a, b by an, bm gives $anbm = ab(b^{-1}nb)m$, which belongs to abN because normality puts $b^{-1}nb$ in N. Associativity descends from G. The projection $\pi(g) = gN$ is a surjective homomorphism with kernel N.

Theorem 10 · First isomorphism theorem
For a homomorphism $f : G \to H$, the map $\overline{f} : G/\ker f \to \operatorname{im} f, g\ker f \mapsto f(g)$, is an isomorphism.

Proof strategy. Do not just count elements. Establish well-definedness, preservation of multiplication, surjectivity, and injectivity separately. **Proof.** Equality of cosets means $b^{-1}a \in \ker f$, hence $f(a) = f(b)$, so the map is well defined. It preserves multiplication since f does. Every element of the image has the form $f(g)$, giving surjectivity. If $\overline{f}(g\ker f) = e$, then $g \in \ker f$, so the quotient kernel is trivial and the map is injective. □

$$\begin{array}{ccc} G & \to f & \operatorname{im} f \\ \downarrow \pi & & \downarrow \mathrm{id} \\ G/\ker f & \to \overline{f} & \operatorname{im} f \end{array}$$

Theorem 11 · Second and third isomorphism theorems
If $H \leq G$ and $N \lhd G$, then HN is a subgroup and $H/(H \cap N) \cong (HN)/N$. If $N \lhd G$, $M \lhd G$, and $N \subseteq M$, then $(G/N)/(M/N) \cong G/M$. Here $\cong$ denotes a group isomorphism.

Proof. Normality allows factors from N to move past factors from H, proving closure and inverse closure of HN. Restrict $G \to G/N$ to H: its kernel is $H \cap N$ and image is HN/N, so the first theorem applies. For the third theorem use $gN \mapsto gM$. It is well defined because $N \subseteq M$, surjective, and its kernel is M/N. Apply the first theorem again. □

Example 11 · Collapsing in stages
Reduction $\mathbb{Z}/12\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ has kernel $\{[0], [4], [8]\}$. Thus quotienting $\mathbb{Z}/12\mathbb{Z}$ by that subgroup gives $\mathbb{Z}/4\mathbb{Z}$. The kernel has three elements and the quotient has four, which agrees with Lagrange.

The correspondence theorem follows from inverse image under π: subgroups of G/N correspond to subgroups of G containing N, by $K \mapsto \pi^{-1}(K)$ and $H \mapsto H/N$. These operations are inverse because π is surjective and H contains the kernel. Normal subgroups correspond to normal subgroups, as conjugation commutes with projection.

Example 12 · Designing a quotient map from the desired answer
To identify $\frac{\frac{\mathbb{Z}}{20}\mathbb{Z}}{\langle [5] \rangle}$, start with reduction $\frac{\mathbb{Z}}{20}\mathbb{Z} \to \frac{\mathbb{Z}}{5}\mathbb{Z}$. It is well defined because five divides twenty. Its kernel is $\{[0], [5], [10], [15]\}$, exactly the named subgroup, so the quotient is $\frac{\mathbb{Z}}{5}\mathbb{Z}$. The order check is $\frac{20}{4} = 5$.

More generally, to show a function on cosets is well defined, suppose two representatives differ by a member of the relevant kernel and show the proposed outputs agree. A formula and a cardinality match alone cannot replace that test. This procedure will be reused for rings, vector spaces, and modules.

Checkpoint 3.2. Why does $gN \mapsto gM$ require $N \subseteq M$?

Takeaways. Most quotient isomorphisms come from a natural map followed by a kernel calculation. State that map before guessing the answer.
Continue with study guide U3 and exercise sheet U3.

4 Unit 4 · Actions and simple groups

4.1 Lecture 7 · Actions, orbits, and stabilizers

Learning goals and prerequisites
Translate a symmetry problem into a group action, compute stabilizers, and prove orbit–stabilizer. Recall cosets and homomorphisms.

Recall and approach. Recall a group action is a homomorphism to permutations of a specified set.

An abstract group becomes visible through what it does to a set. A group can act on several different sets, and each action reveals different information. For example, square symmetries act on vertices, edges, and diagonals; these actions have different stabilizers.

Definition 9 · Group action
A left action of G on X assigns $g \cdot x \in X$ to $g \in G, x \in X$, with $e \cdot x = x$ and $(gh) \cdot x = gh \dot{} x$. The orbit is $Gx = \{g \cdot x : g \in G\}$ and the stabilizer is $G_x = \{g : g \cdot x = x\}$.

For each g, the map $x \mapsto g \cdot x$ is a bijection with inverse the map associated with g^{-1}. Thus an action is equivalently a homomorphism $G \to S_X$. The stabilizer is a subgroup by the subgroup test. Being in the same orbit is an equivalence relation, so the orbits partition X.

Proof strategy. Construct a bijection from cosets to an orbit, and for fixed-point counting count the same incidence set in two orders.

Theorem 12 · Orbit–stabilizer
The map $G/G_x \to Gx, gG_x \mapsto g \cdot x$, is a bijection of sets. For finite G, $|Gx| = |G|/|G_x|$. No normality of G_x is required.

Proof. The images of g and h agree exactly when $h^{-1}g \cdot x = x$, equivalently $h^{-1}g \in G_x$, equivalently $gG_x = hG_x$. This simultaneously proves well-definedness and injectivity. Surjectivity is the definition of the orbit. Count cosets for the final statement. □

Example 13 · The square
The dihedral group D_8 has eight symmetries. It acts transitively on the four vertices. The stabilizer of a vertex consists of the identity and reflection in the diagonal through that vertex, so its order is two and $4 = \frac{8}{2}$. On the two diagonals, a diagonal stabilizer instead has order four.

Theorem 13 · Cayley's theorem
Every group embeds in a permutation group. A finite group of order n embeds in S_n.

Proof. Let G act on itself by left multiplication. The associated homomorphism $\lambda : G \to S_G$ is injective because $\lambda(g) = \mathrm{id}$ implies $g = \lambda(g)(e) = e$. □

Theorem 14 · Counting orbits by fixed points
For finite G, X, the number of orbits is $|G|^{-1} \sum_{g \in G} |X^g|$, where $X^g = \{x : g \cdot x = x\}$.

Proof. Count pairs (g, x) with $g \cdot x = x$. Counting by g gives the numerator; counting by x gives $\sum_x |G_x|$. On one orbit, stabilizers have equal size because $G_{hx} = hG_xh^{-1}$, and orbit–stabilizer makes their total $|G|$. Each orbit contributes once after division. □

Example 14 · Burnside counting on a triangle
Color the three vertices of a triangle with two colors and identify colorings under rotation. There are eight labeled colorings. The identity rotation fixes all eight, while each nonidentity rotation forces all three vertices to share a color and fixes two. The number of orbits is $\frac{8+2+2}{3} = 4$. They can be represented by colorings with zero, one, two, or three vertices of the first color.

This second description checks the count. Dividing eight by three would fail because orbit sizes are not all three: the two monochromatic colorings have larger stabilizers and singleton orbits. The fixed-point average automatically accounts for these different stabilizers.

Checkpoint 4.1. Why is the left multiplication action faithful even when G has a nontrivial center?

Takeaways. Specify the set and the action before computing an orbit or stabilizer. U4 connects these counts to internal group structure.
Continue with study guide U4 and exercise sheet U4.

4.2 Lecture 8 · Conjugacy, the class equation, and simplicity

Learning goals and prerequisites
Use conjugacy classes to restrict normal subgroups and prove simplicity of A_5. Recall permutations, normality, and orbit–stabilizer.

Recall and approach. Recall a conjugation orbit has stabilizer the centralizer, and a normal subgroup contains whole conjugacy classes.

Definition 10 · Center and centralizer
The center is $Z(G) = \{z : zg = gz \quad \text{for every} \quad g\}$. The centralizer of x is $C_{G(x)} = \{g : gx = xg\}$. In the conjugation action $g \cdot x = gxg^{-1}$, the stabilizer is $C_{G(x)}$.

Proof strategy. Partition into conjugacy classes, then combine their allowed unions with subgroup divisibility.

Theorem 15 · Class equation
For finite G, $|G| = |Z(G)| + \sum_i \left[G : C_{G(x_i)}\right]$, where one x_i is chosen from each noncentral conjugacy class. If $|G| = p^n$ with $n \geq 1$ and p prime, then $Z(G)$ is nontrivial.

Proof. Partition G into conjugacy classes. A class has one element precisely when that element is central. Every other class in a group of prime-power order has size a positive power of p greater than one, by orbit–stabilizer and Lagrange. Reducing the class equation modulo p shows p divides $|Z(G)|$. Since the identity is central, the center has at least p elements. □

Definition 11 · Simple group
A nontrivial group is simple if its only normal subgroups are $\{e\}$ and the whole group. A finite abelian simple group is cyclic of prime order: any nonidentity element generates a normal subgroup, and a composite-order cyclic group has a proper nontrivial subgroup.

Theorem 16 · A nonabelian simple group
A_5 is simple.

Proof strategy. Normal subgroups are unions of conjugacy classes containing the identity. Combine class sizes with Lagrange's theorem. **Proof.** The even permutations on five letters have types: identity,

three-cycle, product of two disjoint transpositions, or five-cycle. There are respectively $1, 20, 15, 24$. For a three-cycle, a commuting permutation preserves its three-point support and its two-point complement; the centralizer in S_5 has order $3 \cdot 2$, but its swap on the complement is odd, so the centralizer in A_5 has order three. Thus its A_5 class has size $\frac{60}{3} = 20$. A double transposition fixes a unique point. Its centralizer on the other four points has order eight (choose the image of one point in four ways and then choose whether to interchange its two pairs); exactly half are even, since it contains an odd transposition commuting with the element. Its class therefore has size fifteen. A five-cycle has centralizer its five powers: a commuting permutation is determined by the image of one point. These powers are even, so each A_5 class has size twelve; the twenty-four five-cycles form two classes. A normal subgroup thus has order one plus a subset sum of $20, 15, 12, 12$. The resulting orders are $1, 13, 16, 21, 25, 28, 33, 36, 40, 45, 48, 60$. Only 1 and 60 divide 60. Lagrange finishes the proof. □

This proof explains why a list of possible orders is insufficient: the conjugacy constraint eliminates the otherwise plausible divisors. It also shows that nonabelian simple groups exist without classifying all finite groups.

Watch for this. The assertion "a group has no proper subgroups" is stronger than simplicity and false for A_5. Simplicity excludes proper nontrivial **normal** subgroups.

Example 15 · Why a center constrains the whole group
If $\frac{G}{Z(G)}$ is generated by $gZ(G)$, every element can be written $g^r z$ with z central. Two such elements commute: $(g^r z)(g^s w) = g^{r+s} zw = g^{s+r} wz$. Hence G is abelian. Applied to a group of order p^2, a center of size p would make the quotient cyclic and force the center to be all of G, a contradiction.

This combines a counting argument with a structural lemma. Counting alone permits center sizes p and p^2; understanding a cyclic quotient eliminates the first. In the simplicity proof, the analogous extra information is that normal subgroups are unions of entire classes.

Checkpoint 4.2. Can a group of order p^2 have trivial center?

Takeaways. Actions turn structure into arithmetic. Normality adds a union-of-classes constraint which counting alone cannot see.
Continue with study guide U4 and exercise sheet U4.

5 Unit 5 · Rings and their quotients

5.1 Lecture 9 · Rings, units, matrices, and quaternions

Learning goals and prerequisites
Distinguish additive and multiplicative structure, recognize zero divisors, and compare fields with division rings. Recall matrices and complex arithmetic.

Recall and approach. Recall zero divisors obstruct cancellation and a division ring need not commute.

A ring supports addition and multiplication linked by distributivity. Addition must be an abelian group; multiplication need only be associative. Throughout these notes rings have an identity 1, ring homomorphisms preserve it, and the zero ring is allowed unless explicitly excluded. A field and an integral domain always have $1 \neq 0$.

Definition 12 · Ring vocabulary
A **unit** is an element with a two-sided multiplicative inverse. A nonzero element a is a zero divisor if $ab = 0$ or $ba = 0$ for some nonzero b. An **integral domain** is a commutative nonzero ring without zero divisors. A **field** is a commutative nonzero ring in which every nonzero element is a unit; a **division ring** drops commutativity.

The units $R^\times$ form a group: products of units are units with reversed inverse order. By contrast the nonzero elements need not be closed under multiplication. In $\mathbb{Z}/6\mathbb{Z}$, $[2][3] = [0]$.

Proof strategy. Use injective multiplication in a finite domain; for quaternions, multiply by a conjugate and exploit a positive central norm.

Theorem 17 · Finite domains are fields
Every finite integral domain is a field.

Proof. For nonzero a, multiplication $x \mapsto ax$ is injective: $ax = ay$ implies $a(x - y) = 0$, hence $x = y$. An injective self-map of a finite set is surjective, so some b satisfies $ab = 1$. Commutativity gives $ba = 1$. □

Example 16 · Matrix rings
$M_{n(K)}$ is a ring under matrix operations. For $n \geq 2$ it is noncommutative and has zero divisors. For instance $E_{11}E_{22} = 0$ although neither factor is zero. Its units are precisely the invertible matrices. The scalar matrices form a commutative subring, but nonzero scalar matrices are only some of the units.

Definition 13 · Quaternions
The real quaternion algebra $\mathbb{H}$ has basis $1, i, j, k$, with $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and reversed products negative. For $q = a + bi + cj + dk$, define $\overline{q} = a - bi - cj - dk$.

Associativity can be justified rather than assumed from a multiplication table: write $q = z + wj$ with $z, w \in \mathbb{C}$ and represent it by the complex matrix $\left(\begin{smallmatrix} z & w \\ -\overline{w} & \overline{z} \end{smallmatrix}\right)$. Direct multiplication shows these matrices are closed and reproduce the displayed relations, and the representation is injective. Associativity then follows from matrix multiplication.

Theorem 18 · Quaternion inverses
Every nonzero quaternion has inverse $\frac{\overline{q}}{a^2+b^2+c^2+d^2}$. Thus $\mathbb{H}$ is a division ring but not a field.

Proof. Expanding with the multiplication rules gives $q\overline{q} = \overline{q}q = a^2 + b^2 + c^2 + d^2$. The denominator is a positive real number for nonzero q, hence central and invertible. Noncommutativity follows from $ij = k$ but $ji = -k$. □

Example 17 · Cancellation can fail even for a nonzero scalar
In $\frac{\mathbb{Z}}{12}\mathbb{Z}$, the equation $[4]x = [4]$ has four solutions: $[1], [4], [7], [10]$. Indeed it says 12 divides $4(x - 1)$, or equivalently three divides $x - 1$. Cancelling $[4]$ would incorrectly leave a unique solution. By contrast, multiplication by a unit is a bijection in any ring.

In matrix rings, invertibility is a statement about a whole linear transformation, not about its individual entries. A matrix may have several nonzero entries and still be singular. The quaternion inverse formula succeeds because its norm is a positive real scalar, which is invertible and commutes with every quaternion.

Checkpoint 5.1. Why does the finite-domain proof fail for $\mathbb{Z}$?

Takeaways. Keep the hypotheses "commutative," "finite," and "nonzero" visible. Each has a specific job in the statements above.
Continue with study guide U5 and exercise sheet U5.

5.2 Lecture 10 · Ideals and quotient rings

Learning goals and prerequisites
Construct ideals and ring quotients, prove the ring isomorphism theorem, and recognize prime and maximal ideals.

Recall and approach. Recall ideals absorb multiplication by arbitrary ring elements, not only by their own members.

An additive subgroup alone need not permit multiplication of residue classes. The extra requirement is absorption of multiplication from the ambient ring.

Definition 14 · Ideal
A two-sided ideal I of R is an additive subgroup such that $ra, ar \in I$ for $r \in R, a \in I$. In a commutative ring, the principal ideal (a) consists of all ra.

For $a' = a + i, b' = b + j$, the difference $a'b' - ab = aj + ib + ij$ lies in I. Therefore multiplication of additive cosets is well defined, and R/I is a ring with $(a + I)(b + I) = ab + I$. Its identity is $1 + I$. The projection is a unital ring homomorphism with kernel I.

Proof strategy. Translate quotient properties into ideal conditions and use a map whose kernel expresses the intended relation.

Theorem 19 · First ring isomorphism theorem
For a unital ring homomorphism $f : R \to S$, the kernel is a two-sided ideal and $R/\ker f \cong \operatorname{im} f$ through $r + \ker f \mapsto f(r)$.

Proof. The additive-group kernel is a subgroup. If $f(a) = 0$, then $f(ra) = f(r)f(a) = 0$ and similarly $f(ar) = 0$. The group isomorphism theorem supplies an additive isomorphism; the formula additionally preserves multiplication and identity. □

Definition 15 · Prime and maximal ideals
In a commutative ring, a proper ideal P is prime if $ab \in P$ implies $a \in P$ or $b \in P$. A proper ideal M is maximal if the only ideals containing it are M and R.

Theorem 20 · Quotient criteria
For a commutative ring, P is prime exactly when R/P is a domain; M is maximal exactly when R/M is a field. In particular maximal ideals are prime.

Proof. The prime criterion is the translation of $[a][b] = [0]$ into $ab \in P$, with properness ensuring a nonzero quotient. If M is maximal and $a \notin M$, then $M + (a) = R$, so $1 = m + ra$ and $[r][a] = [1]$. Conversely if the quotient is a field and J properly contains M, choose $a \in J$ outside M. An inverse to $[a]$ gives $1 - ra \in M \subseteq J$, so $1 \in J$ and $J = R$. □

Example 18 · Evaluation as a quotient
Evaluation $K[x] \to K$, $f \mapsto f(c)$, is surjective. Division by $x - c$ shows its kernel is $(x - c)$, so $K[x]/(x - c) \cong K$. Consequently $(x - c)$ is maximal. The quotient isomorphism gives more information than merely saying that c is a root.

Watch for this. For noncommutative rings, left ideals and two-sided ideals differ. Quotient **rings** here require two-sided ideals. The ring $M_2(K)$ has zero divisors even though its only two-sided ideals are zero and itself.

Example 19 · Two similar quotients with different behavior
The quotient $\frac{\mathbb{R}[x]}{x^2+1}$ is a field, while $\frac{\mathbb{R}[x]}{x^2-1}$ is not: the nonzero classes of $x - 1$ and $x + 1$ multiply to zero. They are nonzero because neither degree-one polynomial is a multiple of the degree-two modulus. Thus the second ideal is not prime.

The failure is visible before computing every class. Factoring the modulus produces candidate zero divisors; irreducibility suggests a field, a claim justified over a field coefficient ring in the next unit. This is an example of reading the same relation both as an equation on a root and as an ideal controlling a quotient.

Checkpoint 5.2. Is (0) maximal in $\mathbb{Z}$? Is it prime?

Takeaways. To understand a quotient, first find a useful homomorphism. To understand an ideal, inspect the corresponding quotient.
Continue with study guide U5 and exercise sheet U5.

6 Unit 6 · Polynomials and factorization

6.1 Lecture 11 · Euclidean division and principal ideals

Learning goals and prerequisites
Carry out polynomial division, compute Bézout identities, and prove that Euclidean domains are principal ideal domains.

Recall and approach. Recall a field allows division by every nonzero leading coefficient.

The integer division algorithm has a polynomial analogue because a nonzero leading coefficient in a field can be inverted. Over $\mathbb{Z}$, division by a polynomial with leading coefficient two need not stay inside $\mathbb{Z}[x]$; this is why the coefficient ring matters.

Proof strategy. Cancel the leading term repeatedly for division; minimize the Euclidean measure inside an ideal for the PID proof.

Theorem 21 · Polynomial division
For $f, g \in K[x]$ with K a field and $g \neq 0$, there are unique q, r with $f = qg + r$ and either $r = 0$ or $\deg r < \deg g$.

Proof. If $\deg f < \deg g$, take $q = 0, r = f$. Otherwise subtract the unique scalar multiple of $x^{\deg f - \deg g} g$ cancelling the leading term. The degree decreases, so induction gives existence. If two divisions exist, $(q - q')g = r' - r$. A nonzero left side has degree at least $\deg g$, whereas the right side has smaller degree, a contradiction. □

Definition 16 · Euclidean domain and PID
A Euclidean domain is a domain R with a function $\delta : R \setminus \{0\} \to \mathbb{N}$ such that division $a = qb + r$ is possible with $r = 0$ or $\delta(r) < \delta(b)$. A principal ideal domain (PID) is a domain whose every ideal is generated by one element.

Theorem 22 · Euclidean domains are PIDs
Every Euclidean domain is a PID. In particular $\mathbb{Z}$ and $K[x]$ are PIDs.

Proof. The zero ideal is principal. In a nonzero ideal I, choose $b \neq 0$ minimizing $\delta(b)$. Divide any $a \in I$ by b. The remainder $r = a - qb$ belongs to I; minimality forces it to be zero. Hence every a is a multiple of b, and $I = (b)$. For $\mathbb{Z}$ use absolute value, and for $K[x]$ use degree. □

In a PID the ideal (a, b) equals (d) for a greatest common divisor d. Since $d \in (a, b)$, there are u, v with $d = ua + vb$, a Bézout identity. Conversely $a, b \in (d)$, so d divides both, and any common divisor divides the displayed combination.

Example 20 · An explicit inverse modulo a polynomial
In $\mathbb{Q}[x]$, $x^2 + 1 - x \cdot x = 1$. Thus $-x$ is an inverse to x modulo $x^2 + 1$. The Euclidean algorithm generalizes this: $[f]$ is a unit modulo g exactly when $\gcd(f, g) = 1$.

Theorem 23 · Remainders and roots
For $f \in K[x]$, $f(c) = 0$ if and only if $x - c$ divides f. A nonzero degree-n polynomial has at most n roots in a field.

Proof. Divide by $x - c$; the constant remainder is $f(c)$. If c is a root, factor off $x - c$. Induction on degree bounds the remaining distinct roots, because for $d \neq c$, $0 = f(d) = (d - c)q(d)$ implies $q(d) = 0$ in a field. □

Example 21 · Euclid with a visible Bézout certificate
Over $\mathbb{Q}$, let $f = x^3 - 1$ and $g = x^2 - 1$. Then $f - xg = x - 1$ and $g = (x + 1)(x - 1)$. Hence the monic gcd is $x - 1$, and the first equation is already a Bézout expression for it. The class of g modulo f is therefore not a unit, since the gcd is not one.

Every Euclidean computation has two outputs worth retaining: the gcd and an expression of it in the original inputs. The first records common factors; the second supplies inverses when the gcd is one. Check a computed certificate by expanding it, which detects arithmetic errors independently of the division sequence.

Checkpoint 6.1. Why is $(2, x)$ not principal in $\mathbb{Z}[x]$?

Takeaways. Division gives gcds; gcds give Bézout identities; Bézout identities solve inverse and congruence problems.
Continue with study guide U6 and exercise sheet U6.

6.2 Lecture 12 · Unique factorization and irreducibility

Learning goals and prerequisites
Distinguish irreducible from prime, prove unique factorization in PIDs, and use Gauss's lemma and reduction tests.

Recall and approach. Recall irreducibility forbids factorizations, while primality controls divisibility of products.

Definition 17 · Factorization vocabulary
In a domain, nonzero nonunits p are **irreducible** if $p = ab$ forces a factor to be a unit, and **prime** if $p|ab$ forces $p|a$ or $p|b$. Elements differing by a unit are associates. A unique factorization domain (UFD) admits factorizations into irreducibles, unique up to order and associates.

Theorem 24 · PIDs are UFDs
Every PID is a UFD.

Proof strategy. First prove irreducibles are prime using Bézout. Then rule out infinite failure of factorization using increasing ideals. **Proof.** If irreducible p does not divide a, a gcd of p, a is a unit, so $up + va = 1$. If $p|ab$, multiply this identity by b to obtain $p|b$. For existence, a PID satisfies the ascending chain condition on ideals: the union of an increasing chain is an ideal (d), and d belongs to one member, so the chain stabilizes there. If a nonunit had no finite irreducible factorization, split it into two nonunits and choose a factor still lacking a factorization. Repetition would give an infinite strictly increasing chain of principal ideals, a contradiction. Uniqueness follows because a prime in one factorization divides one factor in another, making them associates; cancel and induct. □

Theorem 25 · Gauss's lemma
For a UFD R, a polynomial is primitive when no prime divides all its coefficients. Products of primitive polynomials are primitive. A primitive polynomial factors over the fraction field only if it factors over R into polynomials of the same positive degrees. Consequently $R[x]$ is a UFD.

Proof. If a prime divided all coefficients of a product of primitive polynomials, reducing modulo that prime would give a product of two nonzero polynomials equal to zero over the domain $R/(p)$, impossible. Every polynomial over the fraction field is a scalar times a primitive polynomial over R, by clearing denominators and removing common prime factors of coefficients. For a factorization of a primitive polynomial, write both factors this way. The product of the primitive parts is primitive, so comparison of the valuation of each prime in all coefficients forces their scalar product to be a unit of R. Absorb that unit into one factor. Factorization over the fraction field, which is a PID polynomial ring, together with factorization of coefficient contents in R, now gives existence and uniqueness in $R[x]$. □

Theorem 26 · Eisenstein's criterion
Let $f = a_n x^n + \ldots + a_0 \in \mathbb{Z}[x]$ be primitive. If a prime p divides every a_i for $i < n$, does not divide a_n, and p^2 does not divide a_0, then f is irreducible over $\mathbb{Q}$.

Proof. By Gauss, a rational factorization yields $f = gh$ in $\mathbb{Z}[x]$ with positive degrees. Reduction modulo p gives a nonzero scalar multiple of x^n, so both reductions are monomials of their original positive degrees (leading coefficients remain nonzero). Both constant terms are divisible by p, contradicting p^2 not dividing a_0. □

Example 22 · Two tests with different scopes
$x^3 - 2$ is Eisenstein at two. The polynomial $x^2 + 1$ is irreducible over $\mathbb{Q}$ because it has no rational root, but it is reducible over $\mathbb{C}$. A root test suffices for degrees two and three; degree four can factor into two quadratics without a root.

Example 23 · Why "no roots" is not a general irreducibility proof
The polynomial $(x^2 + 1)(x^2 + 2)$ has no rational root, since both quadratic factors are positive at every rational input, but it is visibly reducible over $\mathbb{Q}$. Its degree is four. For a polynomial of degree two or three, any nontrivial factorization must include a linear factor, so absence of roots is sufficient there.

When choosing an irreducibility method, first record the degree and coefficient ring. Try a root test in low degree, reduction modulo a prime when the degree survives, or Eisenstein after a useful translation. Failure of one sufficient criterion is not evidence of reducibility; it only means that particular test did not decide the question.

Checkpoint 6.2. Why does an irreducible reduction modulo p imply a primitive integer polynomial is irreducible over $\mathbb{Q}$, provided its degree is preserved?

Takeaways. Always name the coefficient domain. The chain Euclidean domain $\Rightarrow$ PID $\Rightarrow$ UFD has no general reverse implication.
Continue with study guide U6 and exercise sheet U6.

7 Unit 7 · Fields as vector spaces

7.1 Lecture 13 · Fractions, field extensions, and degree

Learning goals and prerequisites
Construct the fraction field of a domain, use its universal property, and prove the tower law. Recall equivalence relations and vector-space bases.

Recall and approach. Recall a domain has valid cancellation by nonzero factors and extension degree is vector-space dimension.

Rational numbers arise by allowing division in the integers. The same construction works in every commutative domain, with the absence of zero divisors providing exactly the cancellations required.

Definition 18 · Fraction field
For a domain R, pairs (a,b) with $b \neq 0$ represent fractions $\frac{a}{b}$. Declare $(a,b) \sim (c,d)$ if $ad = bc$. Define $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ and $\left(\frac{a}{b}\right)\left(\frac{c}{d}\right) = a\frac{c}{bd}$. The resulting field is $\mathrm{Frac}(R)$.

Reflexivity and symmetry are immediate. For transitivity, $ad = bc$ and $cf = de$ imply $d(af - be) = 0$, so $af = be$ since $d \neq 0$. Similar cross multiplication verifies that the operations do not depend on representatives. The denominator product is nonzero. Additive inverses are $-\frac{a}{b}$, and a nonzero fraction has inverse $\frac{b}{a}$. The injection $R \to \mathrm{Frac}(R)$ sends a to $\frac{a}{1}$.

Proof strategy. Verify representative independence for fractions; prove spanning and independence separately for a product basis.

Theorem 27 · Universal property of fractions
If $f : R \to K$ is an injective unital homomorphism into a field, there is a unique extension $\overline{f} : \mathrm{Frac}(R) \to K$ with $\overline{f}\left(\frac{a}{b}\right) = \frac{f(a)}{f(b)}$.

Proof. Injectivity makes each $f(b)$ nonzero. Equality $ad = bc$ gives equality of the proposed values after applying f, proving well-definedness. The addition and multiplication formulas prove preservation of the operations. Any extension must send b^{-1} to $f(b)^{-1}$, so its values are forced. □

Definition 19 · Extension degree
A field extension $\frac{L}{K}$ means K is identified with a subfield of L. Multiplication makes L a vector space over K. Its dimension is the degree $[L : K]$, possibly infinite.

Theorem 28 · Tower law
If $K \subseteq L \subseteq M$ and both $[L : K] = r$ and $[M : L] = s$ are finite, then $[M : K] = rs$.

Proof strategy. Multiply basis elements and prove both spanning and independence. Dimension counting alone is not a proof that the products form a basis. **Proof.** Let $u_1, \ldots, u_r$ be a K-basis of L and $v_1, \ldots, v_s$ an L-basis of M. Expand an element of M in the v_j, then expand its coefficients in the u_i, proving the $u_i v_j$ span. If $\sum_{j(\sum_i a_{ij}u_i)} v_j = 0$ with $a_{ij} \in K$, independence of the v_j gives $\sum_i a_{ij}u_i = 0$ for each j. Independence of the u_i forces every coefficient to vanish. □

Example 24 · A degree-four field
$\mathbb{Q}\left(\sqrt{2},\sqrt{3}\right)$ has basis $1,\sqrt{2},\sqrt{3},\sqrt{6}$ over $\mathbb{Q}$. Indeed $\sqrt{3}$ is not in $\mathbb{Q}\left(\sqrt{2}\right)$: squaring a proposed equality $\sqrt{3}=a+b\sqrt{2}$ forces $ab=0$, and either $a^2=3$ or $2b^2=3$ has no rational solution. The tower law gives degree four.

Example 25 · Degree as a divisibility obstruction
Suppose $\frac{L}{\mathbb{Q}}$ has degree six and contains $\mathbb{Q}\left(\sqrt{2}\right)$. The tower law forces $\left[L:\mathbb{Q}\left(\sqrt{2}\right)\right]=3$. It cannot also contain a subfield of degree four over $\mathbb{Q}$, because four would then divide six. The obstruction concerns an intermediate field inside this particular L; it does not assert that no degree-four field exists elsewhere.

To build a tower basis, first expand in a basis over the intermediate field, then expand those coefficients over the base. For independence, reverse that reasoning: group coefficients by the upper basis, use its independence, and then use independence in the intermediate field. This is more informative than simply multiplying two dimensions.

Checkpoint 7.1. Can a degree-five extension contain a subfield of degree two over the base?

Takeaways. Fields carry both multiplicative structure and linear structure. Degree translates algebraic questions into finite-dimensional reasoning.
Continue with study guide U7 and exercise sheet U7.

7.2 Lecture 14 · Minimal polynomials and adjoining roots

Learning goals and prerequisites
Identify the minimal polynomial, construct a field containing a prescribed algebraic root, and compute in a power basis.

Recall and approach. Recall a minimal polynomial depends on the chosen base field, and a vanishing polynomial may be a multiple of it.

Definition 20 · Algebraic element
An element α in an extension of K is algebraic over K if some nonzero $f\in K[x]$ satisfies $f(\alpha)=0$. Otherwise it is transcendental. For an algebraic element, its minimal polynomial $m_{\alpha,K}$ is the monic polynomial of smallest positive degree vanishing at α.

Proof strategy. Use polynomial division to identify the evaluation kernel, then use irreducibility to make the quotient a field.

Theorem 29 · The minimal-polynomial theorem
The evaluation kernel $\{f:f(\alpha)=0\}$ is generated by $m=m_{\alpha,K}$. The polynomial m is irreducible, and $K[\alpha]=K(\alpha)\cong K[x]/(m)$. If $\deg m=d$, then $1,\alpha,\ldots,\alpha^{d-1}$ is a basis and $[K(\alpha):K]=d$.

Proof. Divide any vanishing f by m; the remainder also vanishes and has smaller degree, so minimality forces it to be zero. If $m=gh$ with both factors of positive smaller degree, then $g(\alpha)h(\alpha)=0$ in a field, forcing one factor to vanish, a contradiction. In the PID $K[x]$, an irreducible generates a maximal ideal: a larger principal ideal would correspond to a proper factor. Thus the quotient is a field. The ring isomorphism theorem identifies it with $K[\alpha]$, which therefore already equals the smallest field $K(\alpha)$. Division gives spanning by the stated powers, and minimality proves their independence. □

Example 26 · Arithmetic in a cubic extension
Let $\alpha^3 = 2$ over $\mathbb{Q}$. Eisenstein proves $x^3 - 2$ irreducible, giving basis $1, \alpha, \alpha^2$. Multiplication is polynomial multiplication followed by reduction $\alpha^3 = 2$. Since $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 1$, the inverse of $\alpha - 1$ is $\alpha^2 + \alpha + 1$.

Theorem 30 · Adjoining a root and splitting a polynomial
Every nonconstant polynomial over a field has a root in some extension and splits into linear factors in a finite extension.

Proof. Choose an irreducible factor p and form the field $K[x]/(p)$. The residue class of x is a root. Over that field, factor off the corresponding linear factor and repeat on the remaining polynomial. Each adjoining step has finite degree, and finitely many steps suffice. The tower law gives a finite final extension. □

If $\frac{L}{K}$ is finite, every element of L is algebraic: $1, \alpha, \ldots, \alpha^n$ are dependent when $n = [L : K]$. Conversely an extension generated by finitely many algebraic elements is finite, by repeated use of the minimal-polynomial theorem and the tower law. Finitely generated is essential: an algebraic extension need not be finite.

Watch for this. A polynomial that vanishes at α need not be its minimal polynomial. Establish monicity and irreducibility, or prove the degree is minimal.

Example 27 · Computing an inverse by reducing a product
In $\mathbb{Q}[\alpha]$ with $\alpha^2 = 2$, multiplication gives $(3 + \alpha)(3 - \alpha) = 7$, hence $(3 + \alpha)^{-1} = \frac{3-\alpha}{7}$. This also follows from a Bézout identity between $x^2 - 2$ and $x + 3$. In higher degree, Euclid replaces the convenient conjugate trick.

To prove a proposed power basis is a basis, division gives spanning by powers below the minimal degree. Any linear dependence of those powers would give a smaller-degree vanishing polynomial, contradicting minimality. Both parts depend on identifying the minimal polynomial, not just finding some relation that happens to vanish.

Checkpoint 7.2. Find the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ and over $\mathbb{Q}\left(\sqrt{2}\right)$.

Takeaways. Quotient rings turn formal roots into actual field elements. The minimal polynomial controls both relations and dimension.
Continue with study guide U7 and exercise sheet U7.

8 Unit 8 · Finite fields and applications

8.1 Lecture 15 · Existence and structure of finite fields

Learning goals and prerequisites
Construct fields of prime-power order and explain why their elements are roots of a single polynomial. Recall splitting fields, degree, and Lagrange.

Recall and approach. Recall the Frobenius binomial identity in prime characteristic and the root bound over a field.

Definition 21 · Characteristic
The characteristic of a nonzero ring is the least positive n with $n \cdot 1 = 0$, or zero when no such n exists. A field of positive characteristic has prime characteristic: if $n = ab$ with $1 < a, b < n$, then $(a \cdot 1)(b \cdot 1) = 0$ contradicts the absence of zero divisors.

A finite field K has characteristic p and contains a copy of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. As a finite-dimensional vector space over this copy, it has p^n elements, one for each coefficient tuple in a basis. Thus the order of every finite field is a prime power.

Proof strategy. Construct all roots in a splitting field, prove they are closed under field operations, and count them using the derivative.

Theorem 31 · Fields of every prime-power order
For each prime p and $n \geq 1$, a field of $q = p^n$ elements exists. It is the set of roots of $x^q - x$ inside a splitting field over $\mathbb{F}_p$.

Proof. Take a finite splitting field by Lecture 14. In characteristic p, the binomial theorem gives $(a + b)^p = a^p + b^p$, because the intermediate binomial coefficients are divisible by p. Iterating gives the same identity for exponent q. Hence roots of $x^q - x$ are closed under addition, negation, and multiplication; if $a \neq 0$ is a root, then $\left(a^{-1}\right)^q = \left(a^q\right)^{-1} = a^{-1}$. The roots contain zero and one, so they form a field. The formal derivative of $x^q - x$ is -1, so no root repeats: a repeated linear factor would divide the derivative. Since the polynomial splits and has degree q, this field has exactly q elements. □

Conversely, in a field of q elements every nonzero element satisfies $a^{q-1} = 1$ by Lagrange in its multiplicative group. Thus all elements are roots of $x^q - x$, and the field is its splitting field over $\mathbb{F}_p$.

Theorem 32 · Uniqueness up to isomorphism
Any two fields of q elements are isomorphic over $\mathbb{F}_p$.

Proof. More generally, an isomorphism of base fields extends to an isomorphism of splitting fields of corresponding polynomials. Adjoin one root at a time: its irreducible minimal polynomial maps to an irreducible polynomial with a root in the other splitting field. The quotient construction gives an extension of the current embedding. Continue through a generating list of roots. The image contains all roots of the target polynomial, since an embedding preserves distinctness and multiplicities, and the target is generated by those roots. Hence the embedding is surjective. Apply this to $x^q - x$. □

Example 28 · A field with four elements
$\mathbb{F}_4 = \mathbb{F}_2[t]/(t^2 + t + 1)$. If $\alpha = [t]$, then $\alpha^2 = \alpha + 1$ and $\alpha^3 = 1$. The elements are $0, 1, \alpha, \alpha + 1$. There is no field of six elements because six is not a prime power.

Example 29 · A complete small-field multiplication check
In $\mathbb{F}_2\frac{[\alpha]}{\alpha^2+\alpha+1}$, the nonzero elements are $1, \alpha, \alpha+1$. The relation gives $\alpha^2 = \alpha+1$, $\alpha(\alpha+1) = 1$, and $(\alpha+1)^2 = \alpha$. Thus every nonzero element has an inverse, while addition satisfies $a + a = 0$ for all elements. This cannot be the ring $\frac{\mathbb{Z}}{4}\mathbb{Z}$, whose element $[2]$ is a nonzero zero divisor.

The characteristic and additive structure give quick diagnostic checks for a claimed finite field. A field of size p^n is an n-dimensional vector space over $\mathbb{F}_p$; its additive group is therefore a product of n cyclic groups of order p, not generally a cyclic group of order p^n.

Checkpoint 8.1. Why does $\mathbb{Z}/4\mathbb{Z}$ fail to be the four-element field?

Takeaways. The notation $\mathbb{F}_q$ denotes a field determined up to isomorphism, not generally the residue ring $\mathbb{Z}/q\mathbb{Z}$.
Continue with study guide U8 and exercise sheet U8.

8.2 Lecture 16 · Multiplicative structure and applications

Learning goals and prerequisites
Prove cyclicity of a finite field's multiplicative group, use interpolation, and connect algebraic constructions to concrete computations.

Recall and approach. Recall finite cyclic group orders and that a nonzero polynomial cannot have too many distinct roots.

Theorem 33 · Finite multiplicative subgroups are cyclic
Every finite subgroup G of the multiplicative group of a field is cyclic. In particular $\mathbb{F}_q^\times$ has a generator of order $q-1$.

Proof strategy. Construct an element whose order is the least common multiple of all element orders. Then use the bound on the number of roots of a polynomial. **Proof.** Let m be that least common multiple. For each prime power ℓ^a dividing m maximally, some element has order divisible by ℓ^a; raising it to the part of its order coprime to ℓ gives an element of order ℓ^a. Elements commute, and the product of commuting elements of pairwise coprime orders has order the product: if a power of the product is one, isolate each factor and use coprimality to show each order divides the exponent. Thus G contains an element of order m. Every element of G is a root of $x^m - 1$, so $|G| \leq m$ by the root bound, while the constructed cyclic subgroup gives $m \leq |G|$. Equality proves the claim. □

Example 30 · Finding a generator
$\mathbb{F}_7^\times$ has order six. The powers of 3 are $3, 2, 6, 4, 5, 1$, so 3 generates it. To test a candidate in a cyclic group of known order N, it suffices to check $g^{\frac{N}{\ell}} \neq 1$ for each prime divisor ℓ of N: any proper divisor order would divide one of these exponents.

Theorem 34 · Lagrange interpolation
For distinct $a_1, \ldots, a_n$ in a field K and prescribed $b_1, \ldots, b_n$, there is a unique polynomial f of degree less than n with $f(a_i) = b_i$.

Proof. Define $L_{i(x)} = \prod_{j \neq i} \frac{x - a_j}{a_i - a_j}$. All denominators are nonzero, and $L_{i(a_k)}$ is one for $k = i$ and zero otherwise. Then $f = \sum_i b_i L_i$ has the required values and degree bound. Two such polynomials have a difference of degree less than n with n roots, so the root bound forces equality. □

This result explains the basic redundancy in polynomial evaluation codes. A degree-less-than-k polynomial is represented by its values at $n \geq k$ distinct field points. Any k correct coordinates determine it.

A nonzero difference of two message polynomials has at most $k-1$ zeros, so two distinct codewords differ in at least $n-k+1$ positions. This is a mathematical distance statement; constructing a decoding algorithm is a further task.

Example 31 · Interpolation over a finite field
Over $\mathbb{F}_5$, the line through $(0,1)$ and $(2,0)$ is $f(x)=1+2x$, because $2^{-1}=3$ and the slope $\frac{0-1}{2}=-3=2$. The field inverse, not ordinary real division, supplies the coefficient.

The Frobenius map $a \mapsto a^p$ is an automorphism of $\mathbb{F}_{p^n}$: it preserves addition and multiplication and is injective, hence surjective on a finite set. Its fixed elements solve $x^p-x=0$, so they are exactly $\mathbb{F}_p$. Advanced Algebra develops this viewpoint further.

Example 32 · Reading the coding bound as a uniqueness argument
Take degree-less-than-three messages over $\mathbb{F}_7$ and evaluate at six distinct points. There are 7^3 messages, and two different messages disagree in at least $6-3+1=4$ positions. If a received word differed from two distinct codewords in at most one position each, those codewords would differ in at most two positions, contradicting the bound. Thus at most one codeword lies within one error of a received word.

This proves uniqueness of correction at that radius, not a procedure for finding the message. The argument separates a structural distance theorem from an algorithmic decoding question and shows exactly how a polynomial root bound becomes information about finite data.

Checkpoint 8.2. How many zeros can the difference of distinct degree-less-than-four messages have?

Takeaways. Finite fields unite the course: groups control nonzero elements, polynomial rings construct the fields, and vector spaces count and encode their elements. Use both mock assessments to test these connections.
Continue with study guide U8 and exercise sheet U8.